

10-2 Find $Z_{EQ}(s)$ in Figure P10-2. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

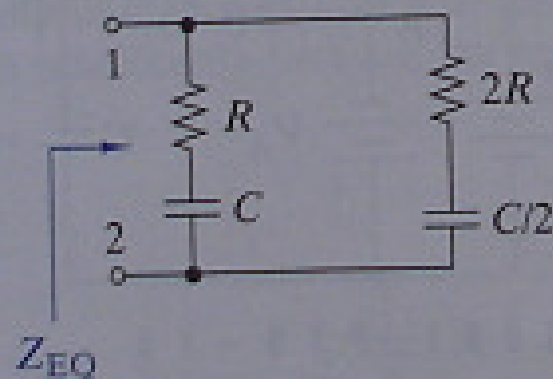


FIGURE P10-2

10-5 Find $Z_{EQ}(s)$ in Figure P10-5. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

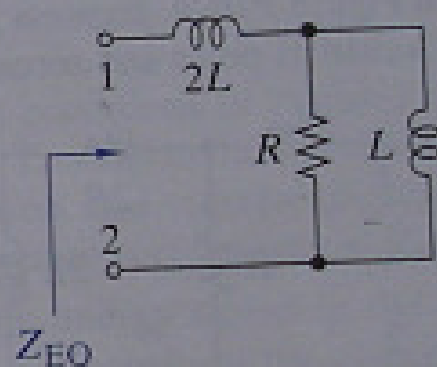


FIGURE P10-5

10-8 Find $Z_{EQ}(s)$ in Figure P10-8. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

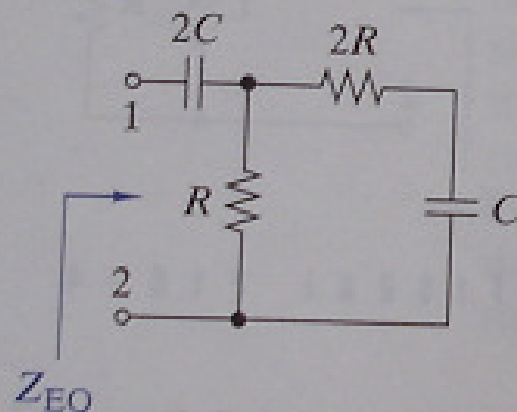


FIGURE P10-8

10-11 The switch in Figure P10-11 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

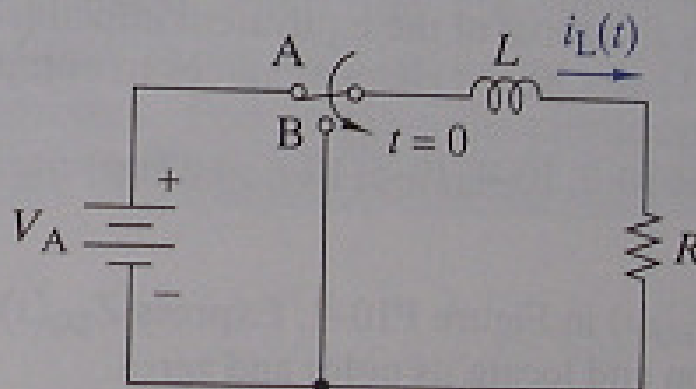


FIGURE P10-11

10-19 The circuit in Figure P10-19 is in the zero state. Find the s -domain relationship between the input $I_1(s)$ and the output $I_R(s)$.

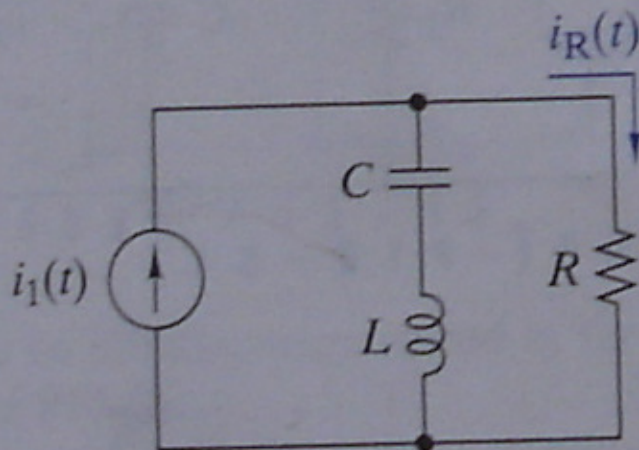


FIGURE P10-19

10-21 The initial conditions in Figure P10-21 are $v_C(0) = V_0$ and $i_L(0) = 0$. Transform the circuit into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_C(s)$.

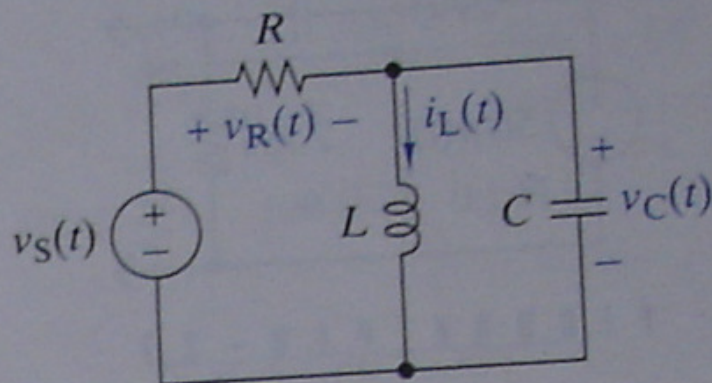


FIGURE P10-21

10-25 The circuit in Figure P10-25 is in the zero state. Use a Thévenin equivalent to find the s -domain relationship between the input $I_S(s)$ and the interface current $I(s)$.

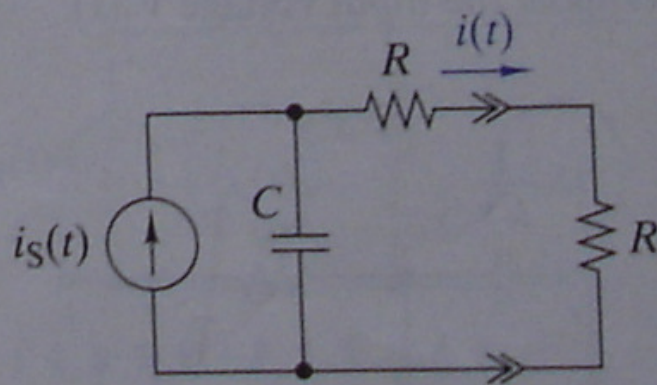


FIGURE P10-25

10-28 There is no initial energy stored in the circuit in Figure P10-28. Transform the circuit into the s domain and use superposition to find $V(s)$. Identify the forced and natural poles in $V(s)$.

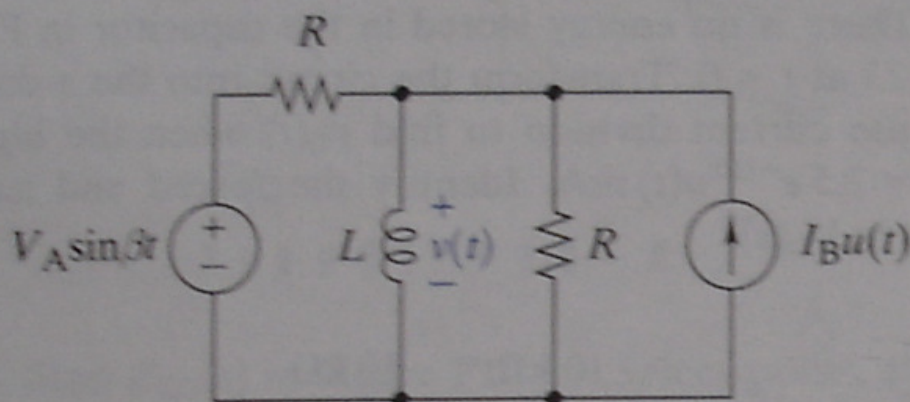


FIGURE P10-28

10-32 There is no initial energy stored in the circuit in Figure P10-31.

- Transform the circuit into the s domain and formulate node-voltage equations.
- Solve these equations for $V_2(s)$ in symbolic form.
- Find $v_2(t)$ for $v_1(t) = 10u(t)$ V, $R_1 = R_2 = 500\ \Omega$, $L = 0.5$ H, and $C = 2\ \mu\text{F}$.

10-37 There is no initial energy stored in the circuit shown in Figure P10-37. Find the zero-state mesh currents $i_A(t)$ and $i_B(t)$ when $v_1(t) = 100u(t)$ V.

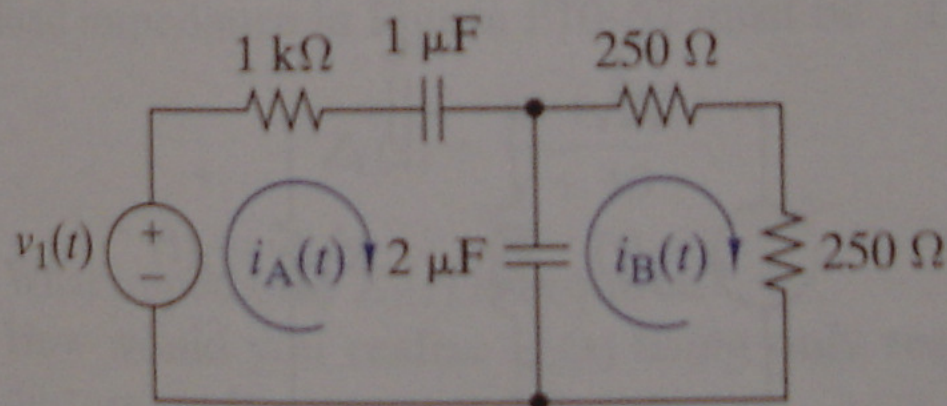


FIGURE P10-37